

Weekly Progress Report — Week 1

July 16 – 22, 2026 · Team Member: Ibrahim Hanafy

Project: AI × Chaos Theory — Time Series Forecasting & ECG Signal Processing

Summary

This week focused on building theoretical foundations across reservoir computing (RC), chaotic systems analysis, and ECG signal preprocessing, while beginning the literature search for the mini review assignment.

Completed

Chaos Theory Fundamentals (Tasks 1 & 2)

Completed simulation of discrete maps (Logistic, Hénon) and continuous chaotic systems (Lorenz, Rössler, Chen). Finished quantitative chaos analysis: Phase Space Reconstruction ($\tau = 17$, $m = 3$), Lyapunov exponent ($\lambda_1 \approx 0.90$ nats/time via ODE), Kolmogorov entropy, correlation dimension ($D_2 \approx 2.06$), and box-counting dimension ($D_0 \approx 1.79$). Full L^AT_EX report and study guide produced.

Reservoir Computing Deep Dive

Studied NG-RC, NG-RC-Ch (Chebyshev polynomial basis), minimal RC, STDM-RC, and physical RC implementations. Prepared a 19-slide presentation covering RC evolution, mathematical framework (ridge regression, feature vectors, Tikhonov regularisation), and applications across chaotic prediction, temperature forecasting, and biomedical signals.

ECG Preprocessing Pipeline (Offline Task 1)

Implemented full preprocessing on the MIT-BIH Arrhythmia Database — 48 records, 7 filtering methods compared (HP Butterworth, Notch, MA, Savitzky–Golay, Chebyshev II, SWT, EMD). Selected optimal pipeline: HP Butterworth (0.5 Hz) → Notch (60 Hz) → SWT (rbio3.9, level 5). Beat segmentation via AAMI mapping yielded 105,039 labelled beats (N/S/V/F/Q classes). Patient-level 80/20 split completed. Output saved as `mitbih_preprocessed.npz`.

Literature Search (Partial)

Found and reviewed 5 papers related to our topics (RC, chaotic time series forecasting, ECG filtering). These include papers from *Chaos*, *Solitons & Fractals*, *Physical Review E*, *Scientific Reports*, and the *Journal of Big Data*.

In Progress / Behind Schedule

- **Mini Review (8 papers in Excel):** Currently at 5/8 papers. Still need 2–3 more papers covering hybrid DL+chaos approaches and chaos-informed ECG classification. The original deadline (July 16) has passed — targeting completion by end of this week.

- **Journal Quality Verification:** Need to cross-check Q-rankings and impact factors on SJR/Scopus for all collected papers and fill remaining Excel columns (Limitations, Comments).

Blockers

None critical. The literature search delay is a time management issue, not a technical one.

Next Week Plan

1. Complete the mini review Excel sheet (8 papers, all columns filled).
2. Begin exploring hybrid approaches: RC + DL, chaos features as DL inputs.
3. Check novelty of potential research directions (NG-RC for ECG? Chaos-informed preprocessing?).
4. Continue notebook development for the forecasting pipeline.